Excel Formulas & Functions Fundamentals

Task 6 · Data Analytics Track · Level 1 – Day 6

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Dataset	Sample Superstore (real dataset) – first 500 of 9,994 rows
Tool	Microsoft Excel
Workbook	Task6_Excel_Formulas_Functions.xlsx
Tabs	Formula_Demo, Data, Lookups, Notes
Formulas	5,000+ live formulas, named ranges, dropdown inputs

1. Objective

Build fluency with the everyday Excel formulas used in real analyst work – lookups, IF logic, conditional sums and counts, and text functions – demonstrated on live data, with notes on when to use each formula.

2. Approach

The Data tab holds 500 real Sample Superstore rows plus 10 calculated columns (text cleaning, VLOOKUP, INDEX/MATCH, nested IF, OR). The Lookups tab holds State→Region and Sub-Category→Category tables derived from the full dataset. The Formula_Demo tab uses named ranges (SalesAmt, OrderRegion, Category, ProfitAmt ...) and yellow input cells so every formula can be re-tested by changing the input. Edge cases (extra spaces, text-vs-number, blank cells, errors) are tested separately.

3. Formula Results (inputs: Region = West, Category = Technology, Segment = Consumer, State = California, Sub-Category = Phones)

Task	Result	Formula
1. SUMIF vs SUMIFS		
SUMIF – sales for region	37,781.64	`SUMIF(OrderRegion,B3,SalesAmt)`
SUMIFS – region + category	15,937.81	`SUMIFS(SalesAmt,OrderRegion,B3,Category,B4)`
SUMIFS – region + category + segment	14,840.56	`SUMIFS(SalesAmt,OrderRegion,B3,Category,B4,Segment,B5)`
SUMIFS – with operator: discount > 30%	1,651.58	`SUMIFS(SalesAmt,OrderRegion,B3,Discount,">0.3")`
SUMIFS – total loss (profit < 0) in region	-1,429.82	`SUMIFS(ProfitAmt,OrderRegion,B3,ProfitAmt,"<0")`
2. COUNTIF / COUNTIFS / AVERAGEIFS		
COUNTIF – rows in region	154	`COUNTIF(OrderRegion,B3)`
COUNTIFS – region + category	32	`COUNTIFS(OrderRegion,B3,Category,B4)`
COUNTIF wildcard – cities starting with "San"	50	`COUNTIF(City,"San*")`
COUNTIF – discount of 40% or more	67	`COUNTIF(Discount,">=0.4")`
AVERAGEIFS – avg sales (region + category)	498.06	`IFERROR(AVERAGEIFS(SalesAmt,OrderRegion,B3,Category,B4),0)`
COUNTBLANK – blank cells in Sales column	0	`COUNTBLANK(SalesAmt)`
3. Lookups: VLOOKUP / XLOOKUP / INDEX-MATCH		
VLOOKUP – region of state B6	West	`VLOOKUP(B6,Lookups!$A$2:$B$50,2,FALSE)`
XLOOKUP – region of state B6	West	`XLOOKUP(B6,StateList,StateRegion,"Not found")`
INDEX/MATCH – category of sub-category B7	Technology	`INDEX(SubCatCat,MATCH(B7,SubCatName,0))`
INDEX/MATCH – LEFT lookup (first sub-category in B4)	Accessories	`INDEX(SubCatName,MATCH(B4,SubCatCat,0))`
XLOOKUP – not found (custom message)	Not found	`XLOOKUP("Atlantis",StateList,StateRegion,"Not found")`
VLOOKUP – not found wrapped in IFERROR	Not found	`IFERROR(VLOOKUP("Atlantis",Lookups!$A$2:$B$50,2,FALSE),"Not found")`
4. IF logic		
IF – is region profitable?	Profitable	`IF(SUMIFS(ProfitAmt,OrderRegion,B3)>0,"Profitable","Loss-making")`
Nested IF – size of SUMIFS total	Low	`IF(B11>=50000,"High",IF(B11>=20000,"Medium","Low"))`
IF + AND – segment health check	Strong	`IF(AND(B17>50,B21>200),"Strong","Review")`

Task	Result	Formula
5. Text functions		
Raw city input	los ANGELES	None
Raw location string input	Henderson|Kentucky|42420	None
TRIM – remove extra spaces	los ANGELES	TRIM(B38)
PROPER(TRIM) – clean city	Los Angeles	PROPER(TRIM(B38))
UPPER	LOS ANGELES	UPPER(TRIM(B38))
LEN – raw vs clean length	14 vs 11	LEN(B38)&" vs "&LEN(TRIM(B38))
LEFT – first 3 letters of state	CAL	UPPER(LEFT(B6,3))
FIND – position of first |	10	FIND("|",B39)
MID + FIND – city part of string	Henderson	MID(B39,1,FIND("|",B39)-1)
SUBSTITUTE – replace | with comma	Henderson, Kentucky, 42420	SUBSTITUTE(B39,"|",", ")
& (concatenate) – label	Region West – Technology (Consumer)	"Region "&B3&" – "&B4&" ("&B5&")"
TEXT – format number	37,782	TEXT(SUMIFS(SalesAmt,OrderRegion,B3),"#,##0")
6. Edge-case tests (blank cells, text-vs-number, spaces, errors)		
Test input: state with spaces	California	None
Test input: postal code stored as TEXT	42420	None
Test input: blank cell (leave empty)		
VLOOKUP without TRIM -> fails	Not found (spaces!)	IFERROR(VLOOKUP(B52,Lookups!$A$2:$B$50,2,FALSE),"Not found (spaces!)")
VLOOKUP with TRIM -> works	West	IFERROR(VLOOKUP(TRIM(B52),Lookups!$A$2:$B$50,2,FALSE),"Not found")
ISNUMBER on text postal code	False	ISNUMBER(B53)
COUNTIF coerces: text criteria still matches numbers	2	COUNTIF(Data!$G$2:$G$501,B53)
VALUE converts text to number	42420	VALUE(B53)+0
Blank cell check	Blank cell	IF(B54="","Blank cell","Has value")
N() turns blank into 0	0	N(B54)
Division by zero handled	Div by zero handled	IFERROR(1/0,"Div by zero handled")

Note: the two XLOOKUP formulas need Excel 365/2021+ or Google Sheets.

4. When to Use Each Formula

Function	Use it when…	Watch out for
SUMIF	Total with ONE condition (e.g. sales for one region).	Order: SUMIF(range, criteria, sum_range).
SUMIFS	Total with TWO or MORE conditions (region + category + discount).	Sum_range comes FIRST in SUMIFS, last in SUMIF.
COUNTIF / COUNTIFS	Count rows matching one / many conditions; wildcards like "San*"; operators like ">=0.4".	Numbers stored as text may not match.
AVERAGEIFS	Average of values meeting conditions (avg sale per region).	Gives a divide-by-zero error if nothing matches – wrap in IFERROR.
VLOOKUP	Quick lookup when the key is the LEFT-most column of the table.	Use FALSE for exact match; breaks if columns are inserted; cannot look left.
XLOOKUP	Modern replacement: any direction, exact match by default, built-in not-found message.	Needs Excel 365/2021+ or Google Sheets.
INDEX/MATCH	Look left, support older Excel, or stay safe when columns change.	Use MATCH(...,0) for exact match.
IF / nested IF	Return different results by condition (profit status, discount band).	Keep nesting shallow; use a lookup table for many tiers.
AND / OR	Combine several conditions inside IF.	Each condition must be a full test.
IFERROR	Show a friendly value instead of an error.	Do not hide real errors – use only where an error is expected.
TRIM / PROPER / UPPER	Clean messy text before lookups and grouping.	TRIM does not remove non-breaking spaces.
LEFT / MID / FIND / LEN	Pull parts of text (state code, split strings).	Wrap FIND in IFERROR when the character may be missing.
SUBSTITUTE	Replace specific text.	Case-sensitive.
VALUE / ISNUMBER / N	Fix text-vs-number mismatches and blanks.	Check with ISNUMBER first.
Named ranges	Make formulas readable (SUMIFS(SalesAmt,OrderRegion,...)).	Extend ranges when data grows, or use an Excel Table.

5. Interview Questions

What is the difference between SUMIF and SUMIFS?
SUMIF handles one condition; SUMIFS handles many. SUMIFS takes the sum_range first, SUMIF takes it last.

How does XLOOKUP improve on VLOOKUP?
It looks left or right, uses exact match by default, needs no column index number (safe if columns change), has a built-in not-found value, and can return multiple columns.

When would you use INDEX/MATCH instead of VLOOKUP?
When the return column is to the left of the key, when you need older Excel without XLOOKUP, or when columns may be inserted or deleted.

6. Conclusion

The workbook demonstrates every required function on real data, handles edge cases such as extra spaces, text-formatted numbers and blank cells, and documents when each formula is the right choice.